

HOLOD, Romania

WMO: 151170

Lat: 46.7886N

Lon: 22.1122E

Elev: 165

StdP: 99.36

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-8.4	-13.3	1.2	-9.9	-10.5	1.6	-7.5	10.4	3.1	8.3	2.5	2.0	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	33.2	21.9	31.4	21.6	29.6	21.0	23.4	30.4	22.5	29.1	21.8	27.9	2.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.4	26.0	20.6	15.6	25.0	19.9	14.9	24.1	70.3	30.3	67.1	29.3	64.3	27.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.3	5.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-0.4	3.1	6.0	11.4	16.4	20.3	21.3	21.9	17.0	11.6	7.2	1.5
	DBStd	8.66	4.81	4.51	4.20	4.65	3.35	3.21	3.11	3.25	4.23	4.13	4.68	4.57
	HDD10.0	1099	321	194	134	40	2	0	0	0	1	32	109	265
	HDD18.3	2846	579	426	382	208	76	17	8	6	76	211	335	522
	CDD10.0	1644	0	2	11	83	202	308	351	371	211	82	25	1
	CDD18.3	350	0	0	0	2	18	75	101	118	35	2	1	0
	CDH23.3	3592	0	0	0	38	212	707	996	1244	359	36	1	0
CDH26.7	1294	0	0	0	2	33	221	378	543	116	1	0	0	
Wind	WSAvg	2.4	2.0	2.5	3.0	2.9	2.5	2.3	2.2	2.1	2.2	2.2	2.4	2.1
Precipitation	PrecAvg	708	44	39	42	57	63	86	82	55	66	60	55	60
	PrecMax	936	77	113	164	106	138	189	153	115	174	139	101	142
	PrecMin	431	19	2	7	10	16	14	30	1	7	3	0	0
	PrecStd	141	18	24	36	24	31	38	39	33	41	36	24	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	15.9	21.0	26.6	29.2	33.0	34.7	35.9	32.6	26.0	21.3	14.7
		MCWB	9.5	10.1	13.3	16.5	19.2	23.2	22.1	22.1	21.3	17.0	13.9	10.0
	2%	DB	9.6	13.4	17.8	24.3	27.5	31.1	32.7	33.7	29.5	24.3	19.0	11.3
		MCWB	7.4	9.6	11.2	15.6	19.3	22.5	22.0	21.6	19.8	16.5	13.5	8.7
	5%	DB	7.5	11.6	15.0	22.1	25.8	29.4	30.5	31.8	27.6	21.9	16.5	9.1
		MCWB	6.1	8.2	9.7	14.5	17.9	21.7	21.3	21.3	18.8	15.7	12.4	7.5
	10%	DB	6.1	9.7	12.8	19.9	23.9	27.4	28.5	29.6	25.0	19.1	14.2	7.1
		MCWB	5.1	7.4	8.3	13.3	17.5	20.7	20.4	20.9	18.1	14.7	11.4	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	11.4	14.2	17.9	21.0	24.7	23.8	24.2	23.5	19.4	15.3	10.5
		MCDB	12.7	14.1	20.1	24.0	26.3	31.4	31.4	31.8	30.2	23.3	20.1	13.4
	2%	WB	8.0	10.1	12.3	16.3	20.0	23.3	22.8	23.1	21.2	17.7	13.9	9.2
		MCDB	9.4	12.4	16.8	22.9	25.5	29.5	29.7	30.7	25.6	21.6	18.0	11.2
	5%	WB	6.5	8.9	10.6	15.2	19.1	22.3	22.1	22.2	20.0	16.3	13.0	7.5
		MCDB	7.6	11.2	14.5	21.0	24.5	28.2	28.1	29.4	24.7	20.4	16.7	9.0
	10%	WB	5.0	7.6	9.2	13.9	18.1	21.4	21.4	21.5	18.9	15.1	11.6	6.2
		MCDB	6.1	9.6	12.3	19.3	22.7	26.7	27.0	28.2	23.6	18.6	14.2	7.3
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.8	9.9	11.6	11.3	11.5	12.0	12.7	11.2	11.1	8.3	5.6
		MCDBR	8.5	10.8	13.2	14.8	14.8	14.0	14.8	15.9	15.3	15.9	11.6	8.3
		MCWBR	6.2	7.1	7.7	7.7	7.2	6.1	5.5	5.4	6.2	8.7	6.5	5.8
	5% WB	MCDBR	7.7	9.6	12.3	13.5	12.6	12.6	12.6	13.8	12.4	13.3	10.3	7.8
		MCWBR	5.8	6.4	7.5	7.2	6.6	6.1	5.2	5.1	5.6	7.8	6.0	5.6
Clear-Sky Solar Irradiance	taub	0.327	0.355	0.395	0.448	0.432	0.431	0.448	0.449	0.417	0.381	0.353	0.328	
	taud	2.331	2.279	2.212	2.084	2.195	2.235	2.195	2.189	2.245	2.332	2.358	2.363	
	Ebn at Noon	747	799	819	805	836	838	816	797	787	760	709	700	
	Edh at Noon	76	98	121	151	140	135	139	135	116	92	74	67	
All-Sky Solar Radiation	RadAvg	1.12	1.95	3.16	4.47	5.42	5.94	5.83	5.33	3.75	2.51	1.42	0.87	
	RadStd	0.15	0.28	0.45	0.45	0.52	0.45	0.50	0.50	0.42	0.29	0.17	0.12	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (64 neighbors)		+0.70	N/A	N/A	+1.00	N/A	N/A	-127	-191	+112	+57	

Nomenclature: See separate page